Module 1: Fundamentals of Cloud Computing

1. **Definition:** Cloud computing is the delivery of on-demand computing services over the internet.
2. **Core Resources:** These services include servers, storage, databases, networking, analytics, and software.
3. **Utility Model:** It operates on a pay-as-you-go economic model, resembling a public utility.
4. **NIST Standard:** The National Institute of Standards and Technology defines its baseline characteristics.
5. **Evolution:** It evolved by merging mainframes, cluster computing, grid computing, and distributed networks.
6. **On-Demand Self-Service:** Users can provision computing capabilities automatically without human intervention from the provider.
7. **Broad Network Access:** Services are available over the network and accessed through standard mechanisms (mobiles, laptops).
8. **Resource Pooling:** The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model.
9. **Location Independence:** Customers generally have no control or knowledge over the exact location of the provided resources.
10. **Rapid Elasticity:** Resources can be elastically provisioned and released to scale rapidly outward and inward with demand.
11. **Measured Service:** Cloud systems automatically control and optimize resource use by leveraging a metering capability.
12. **Capital Expenditure (CapEx):** Traditional IT requires high upfront costs for physical hardware and data centers.
13. **Operational Expenditure (OpEx):** Cloud shifts costs to ongoing operational expenses based on actual usage.
14. **Total Cost of Ownership (TCO):** Cloud drastically reduces TCO by eliminating hardware maintenance, cooling, and real estate costs.
15. **Data Centers:** The physical foundation of the cloud, housing thousands of networked servers globally.
16. **Front-End Architecture:** Comprises the client infrastructure, user interfaces, and applications used to access the cloud.

17. **Back-End Architecture:** Refers to the cloud service provider's resources, including storage, virtual machines, and security mechanisms.
18. **Middleware:** Specialized software layers that manage connectivity, database integration, and resource orchestration.
19. **Service Level Agreements (SLAs):** Formal contracts defining the committed uptime, performance, and remedies between provider and user.
20. **High Availability:** Architecture designed to ensure system operational continuity and minimize downtime. [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11]

Module 2: Cloud Service Models

21. **SPI Framework:** The foundational service stack consisting of IaaS, PaaS, and SaaS.
22. **IaaS Definition:** Infrastructure as a Service provides fundamental computing, storage, and networking resources.
23. **IaaS Control:** Users manage operating systems, storage, and deployed applications, but not the physical infrastructure.
24. **IaaS Examples:** Amazon EC2, Microsoft Azure VMs, and Google Compute Engine.
25. **PaaS Definition:** Platform as a Service provides a hardware and software framework for building applications.
26. **PaaS Efficiency:** Developers focus on coding and data management without worrying about managing server OS or patching.
27. **PaaS Examples:** Google App Engine, AWS Elastic Beanstalk, and Heroku.
28. **SaaS Definition:** Software as a Service delivers a complete, fully functional software application over the web.
29. **SaaS Accessibility:** Entirely managed by the vendor and typically accessed via web browsers or light desktop clients.
30. **SaaS Examples:** Salesforce CRM, Google Workspace (Gmail), Slack, and Zoom.
31. **XaaS:** "Anything as a Service" covers modern extensions like Storage-as-a-Service and Database-as-a-Service.
32. **CaaS:** Container as a Service manages container deployment, scaling, and orchestration pipelines.

33. **FaaS:** Function as a Service allows executing modular pieces of code in response to specific system events.
34. **Serverless Computing:** A FaaS execution model where the cloud provider completely manages machine provisioning and idle time.
35. **Shared Responsibility Model:** Dictates security ownership; providers secure the infrastructure, while users secure data within it. [1, 2, 3, 4, 5, 6, 7]

Module 3: Deployment Models

36. **Public Cloud:** Resources are owned and operated by a third-party provider and shared across multiple tenants over the internet.
37. **Public Cloud Pro:** Offers maximum scalability, flexibility, and zero localized hardware capital investment.
38. **Private Cloud:** Infrastructure is provisioned for the exclusive use of a single organization.
39. **Private Cloud Pro:** Provides high customization, control, data sovereignty, and robust security for strict compliance.
40. **Hybrid Cloud:** Combines public and private infrastructures, bound together by standardized technology.
41. **Hybrid Cloud Benefit:** Allows data and applications to move seamlessly between environments, balancing cost and control.
42. **Cloud Bursting:** An application deployment model where private cloud workloads spill into a public cloud during peak demands.
43. **Community Cloud:** Infrastructure shared by several organizations with common concerns (e.g., security compliance, jurisdiction).
44. **Multi-Cloud:** The strategic use of multiple distinct public cloud providers to prevent vendor lock-in and optimize features.
45. **Edge Computing:** Decentralized framework bringing computing power and data storage closer to the source of data generation. [1, 2, 3, 4, 5, 6, 7]

Module 4: Virtualization Technologies

46. **Virtualization Definition:** The core technology enabling cloud by abstracting physical hardware into virtual machines (VMs).
47. **Hypervisor / VMM:** Virtual Machine Monitor software that creates, runs, and isolates virtual environments on hardware.

48. **Type 1 Hypervisor:** Bare-metal hypervisors running directly on the physical host hardware (e.g., VMware ESXi, KVM).
49. **Type 2 Hypervisor:** Hosted hypervisors running as a software layer on top of a host operating system (e.g., VirtualBox).
50. **Full Virtualization:** The guest OS is unmodified and unaware it is virtualized, executing through hypervisor translation.
51. **Paravirtualization:** The guest OS is explicitly modified to communicate via hypercalls directly to the hypervisor (e.g., Xen).
52. **CPU Virtualization:** Multiplexes a single physical processor among several independent virtual execution contexts.
53. **Memory Virtualization:** Manages physical RAM abstraction by mapping guest physical memory addresses to actual host memory.
54. **I/O Virtualization:** Manages access sharing of physical network adapters and storage controllers across virtual instances.
55. **Virtual Machines (VMs):** Emulated computer systems running an isolated operating system stack over shared hardware.
56. **Containers:** OS-level virtualization sharing the host kernel, making them lightweight alternatives to VMs.
57. **Docker:** A highly popular open-source platform used to automate application deployment within lightweight containers.
58. **Kubernetes (K8s):** The industry-standard orchestration system used for automating container scaling and management.
59. **Live Migration:** Moving a running virtual machine from one physical host to another with near-zero application downtime.
60. **Server Consolidation:** Combining workloads from multiple underutilized physical servers into virtual instances on a single host. [1, 2, 3, 4]

Module 5: Cloud Storage & Networking

61. **Object Storage:** Unstructured data storage managed as discrete objects with metadata, ideal for files and images (e.g., AWS S3).
62. **Block Storage:** Divides data into raw volumes, acting like local hard drives, ideal for transactional databases (e.g., AWS EBS).
63. **File Storage:** Hierarchical data storage accessed via network file sharing protocols (NFS/SMB) for concurrent access.

64. **Ephemeral Storage:** Temporary disk storage tied directly to the lifetime of a specific running virtual instance.
65. **Virtual Private Cloud (VPC):** A logically isolated virtual network dedicated entirely to a user's cloud account.
66. **Subnetting:** Segmenting a VPC IP address range into smaller, manageable isolated sections for security zoning.
67. **Load Balancing:** Automatically distributes incoming network traffic across multiple healthy target servers to maximize availability.
68. **Content Delivery Network (CDN):** Geographically distributed proxy servers that cache media assets closer to end users.
69. **DNS Services:** Cloud routing registries translating human-readable web domains into numeric IP directions.
70. **VPN Gateway:** Provides encrypted network tunnels connecting on-premises corporate infrastructure safely to cloud networks. [1, 2, 3]

Module 6: Resource Management & Scaling

71. **Horizontal Scaling (Scale Out/In):** Adjusting capacity by adding or removing compute instances to a workload pool.
72. **Vertical Scaling (Scale Up/Down):** Adjusting capacity by adding or removing CPU, RAM, or storage resources to a single instance.
73. **Auto-Scaling:** Automatic dynamic configuration adjusting active resource pools based on real-time metric thresholds.
74. **Resource Elasticity:** The exact matching of resource supply to fluctuating demand curves in real time.
75. **Over-Provisioning:** Allocating more resources than required, resulting in unnecessary operational costs.
76. **Under-Provisioning:** Allocating too few resources, causing performance degradation or application outages.
77. **Load Metric Tracking:** Monitoring utilization rates of CPU, memory space, network input/output, and request queues.
78. **Fault Tolerance:** System resilience mechanisms allowing applications to continue operations despite underlying hardware failures.
79. **Disaster Recovery (DR):** Comprehensive data strategies using geo-replication to recover business operations after catastrophic failures.

80. **Recovery Time Objective (RTO):** The maximum targeted duration of time allowed to restore systems after an outage.
81. **Recovery Point Objective (RPO):** The maximum targeted age of data that can be lost due to a disruptive event. [1, 2, 3, 4, 5, 6, 7]

Module 7: Cloud Security & Governance

82. **Identity and Access Management (IAM):** Framework managing user digital identities and granular resource permission privileges.
83. **Data Encryption at Rest:** Protecting stored files and databases using cryptographic ciphers (AES-256).
84. **Data Encryption in Transit:** Safeguarding data packets moving across networks using TLS/SSL cryptographic protocols.
85. **Multi-Tenancy Risks:** Security issues arising from logical errors allowing cross-tenant data leaks over shared hosts.
86. **Cloud Security Alliance (CSA):** Leading global organization defining best practices to secure cloud environments.
87. **Distributed Denial of Service (DDoS) Protection:** Cloud filters mitigating massive, malicious bot network traffic spikes.
88. **Compliance Frameworks:** Regulatory standards governing cloud architecture, such as GDPR, HIPAA, and PCI-DSS.
89. **Cloud Governance:** The overarching rule policies, budget guardrails, and allocation controls established for enterprise usage.
90. **Data Sovereignty:** Legal requirements dictating that digital data must remain physically stored within the nation of origin. [1, 2, 3, 4, 5, 6]

Module 8: Distributed Frameworks, Migration, & Trends

91. **Big Data Systems:** Large-scale distributed processing architectures optimized for unstructured enterprise datasets.
92. **Hadoop Framework:** Open-source framework enabling distributed batch processing across massive data clusters.
93. **HDFS:** Hadoop Distributed File System, providing highly fault-tolerant storage across cluster machines.
94. **MapReduce:** Software paradigm filtering and processing immense data sets in parallel across distributed nodes.

95. **Cloud Migration:** The structured process of relocating digital assets, databases, and apps to cloud environments.
96. **Lift and Shift (Rehosting):** Migrating applications directly to the cloud without modifying their base code architecture.
97. **Refactoring:** Overhauling and rewriting application code to fully exploit cloud-native features and microservices.
98. **FinOps:** An operational practice combining finance and tech to optimize spending efficiency in cloud architectures.
99. **Green Computing:** Implementation of energy-efficient hardware, dynamic cooling, and dynamic power management in data centers.
100. **Future AI Convergence:** Rapid growth of specialized GPU cloud platforms to train and run massive artificial intelligence models.